

A thick black L-shaped frame is positioned on the left and bottom edges of the slide, framing the central text.

A TWO DOF SELF ADAPTIVE BIONIC ANKLE

Shanghai High School International Division

Justin Guo

Background Significance and Proposal

- For a human robot, any part of the system needs careful consideration. For robots that focuses more on movement ability, including needing the ability to move through rough terrain and great vertical distance. These robots may take the role to traverse through mountain area, rescue people in a building, or going on a mission to mars.
- **Proposal: A good leg, or more specific, a good ankle will help the robots to traverse better in complex environments.**
- Which will include:
 - an additional motor to control not only the leg's dorsiflexion and plantar flexion, but also **inversion and eversion movement** of an ankle.
 - **a self adaptive program** for it to aid walking using the combination of IMU and laser distance sensors.



Theoretical Design

- The bionic ankle mimics a human ankle in many ways, including joint, muscles, ligaments, and bones.
- The torque of the ankle can be calculated as

$$T_{motor} \times \frac{d1+d2}{r}$$
- Where d1 is the distance of the front string to the ankle, d2 is the distance of the back string to the ankle, T is the torque of the motor, and r is the radius of the axle.
- The height of the whole ankle should be 500 mm, which is human's knee down length.
- The maximum angle of ankle should generally be close to the actual numbers



Figure 5: Side view of the Human Ankle's Bone System

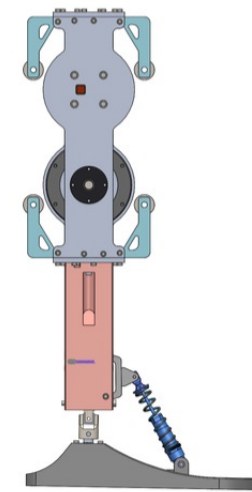


Figure 6: Side view of the Bionic Ankle

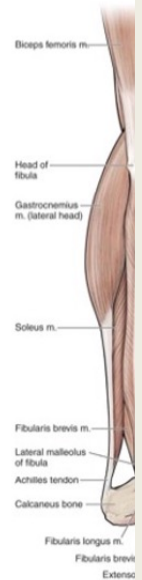
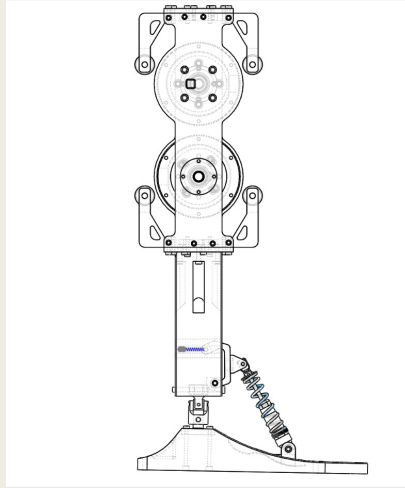
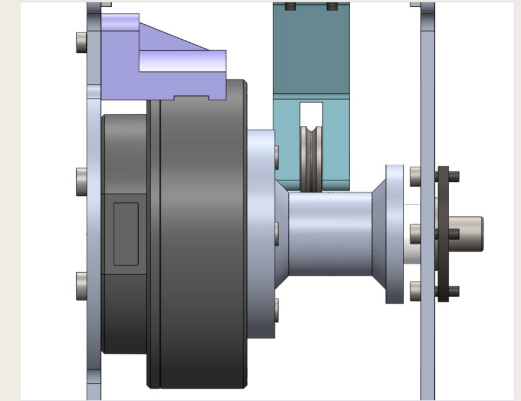


Figure 7: Side view of the Human Ankle's Muscle System

Mechanical Ankle	Human Ankle	Function
Spring System	Articular Cartilage, Ligaments	Absorb shock, provide connection support
3D-printed Leg	Bones: Fibula, Tibia	Support the body and the drive system
Universal Joint(Block)	Bone: Talus	Provide surface to the two joints
Universal Joint(Upper Yolk)	Upper Ankle Joint	Allows dorsiflexion and plantar flexion
Universal Joint(Lower Yolk)	Lower Ankle Joint	Allows inversion and eversion
Drive System	Muscles: fibularis longus, tibialis anterior	Controls the movement of the ankle
3D-printed Foot	The whole foot	Provides a large surface for balancing the



Part	Type	Numbers
MS9015	bought	2
TTL - RS485	bought	1
steel axle	cut	2
wider axle	3D printed	2
damping	3D printed	2
track rollers	bought	2
flange bearings	bought	2
flange bearings cover	laser cutting	2
guide wheels	bought	4
guide wheels supporter	3D printed	4
side board	customized	2
top board	customized	2
bottom board	customized	2
supporting leg	3D printed	1
springs	bought	2
tension springs (different statistics)	bought	6
tension spring connector	customized	2
spring connector	bought	2
universal joint	bought	1
universal joint connector	customized	1
foot	3D printed	1
JY62	bought	1
VL53L0X	bought	4
TCA9548A	bought	1
esp32 board	bought	3



Structural Design

- Use Solidworks to plan out the whole structure and record the parts needed to build the whole structure.
- The whole leg's proportions close to human statistics. (The height of the bionic ankle is 500mm).

HARDWARE DESIGN

For IMU, distance sensor, and motor, I choose the most accurate one to use.

Name	Type	Image	Usage
JY62	IMU		Give the pitch, roll, and yaw of the foot
VL53L0X	ToF sensor		Give the distance to the nearest object
TCA9548A	switch		Transmits multiple devices' signal
esp32	control board		Controls other devices
MS9015	motor		Turns the axle
24V battery	battery		Provides voltage
TTL-RS485	communication conversion module		Converts different communication protocols

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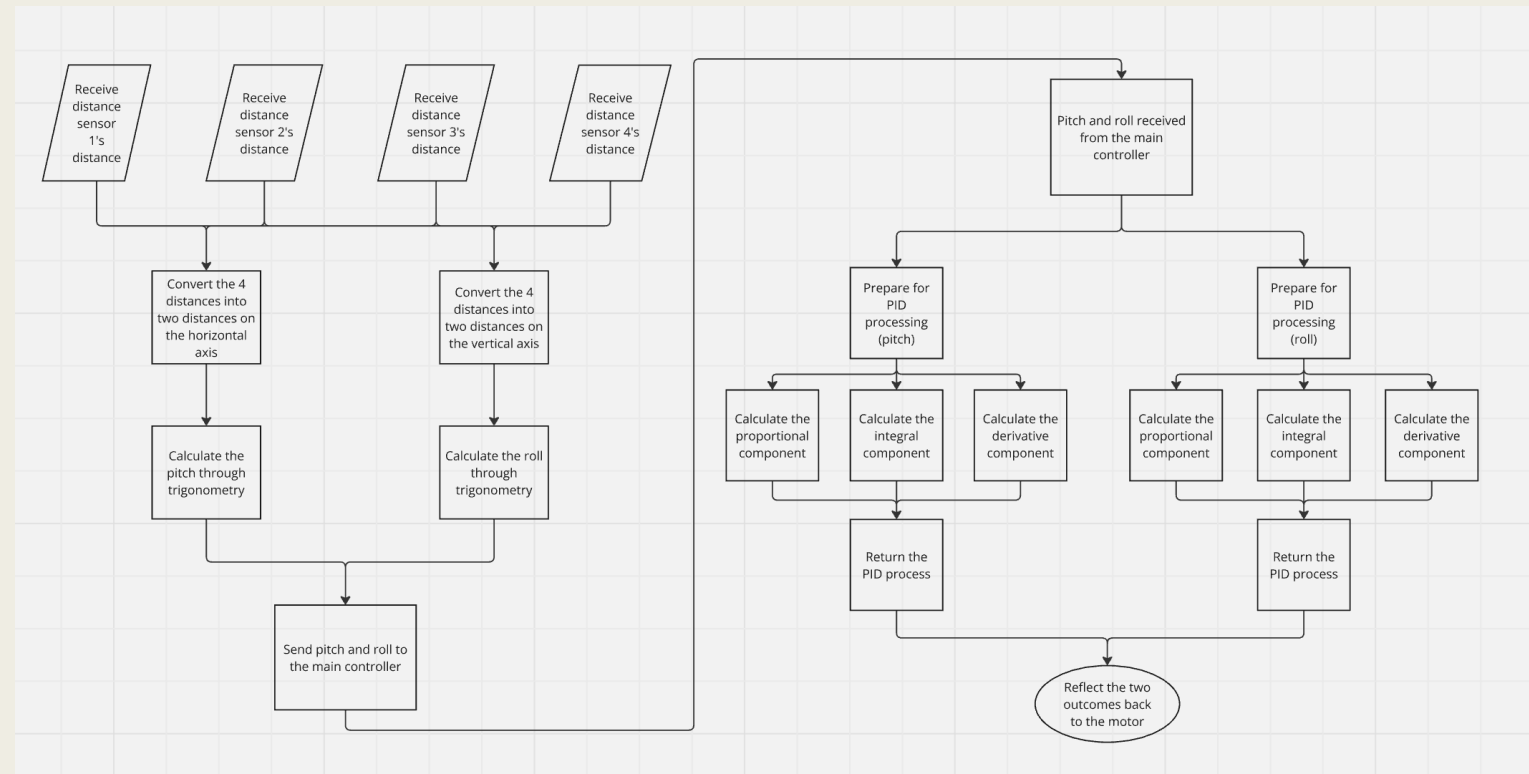
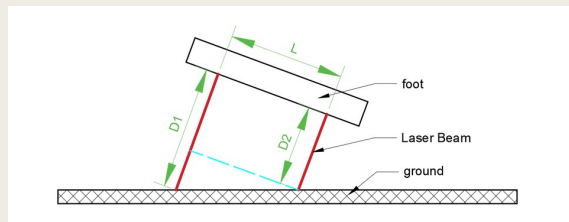
float PID_control (float now_date, float set_date)
{
    float err = set_date - now_date;
    err_all = err + err_all;

    if (err_all > 1000) err_all = 1000;
    if (err_all < -1000) err_all = -1000;
    float P_OUT = kp * err;
    float I_OUT = ki * err_all;
    float D_OUT = kd * (err - last_err);
    last_err = err;
    float all_out = P_OUT + I_OUT + D_OUT;
    if (all_out > 1200) all_out = 1200;
    if (all_out < -1200) all_out = -1200;
    return all_out;
}

float PID_control2 (float now_date2, float set_date2)
{
    ...
}

void PID_total (float roll, float pitch, float set_pitch, float set_roll) {
    roll_PID_angle = PID_control(roll, set_roll);
    pitch_PID_angle = PID_control2(pitch, set_pitch);
    angle_increased_send(1, -roll_PID_angle + pitch_PID_angle);
    delay(10);
    angle_increased_send(2, roll_PID_angle + pitch_PID_angle);
    delay(10);
}

```



SOFTWARE PROGRAMMING

Maximum Angle Experiment

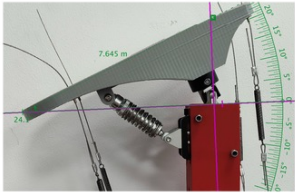


Figure 36: Dorsiflexion

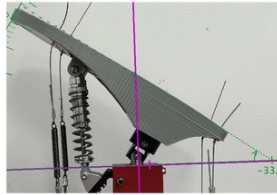


Figure 37: Plantar Flexion

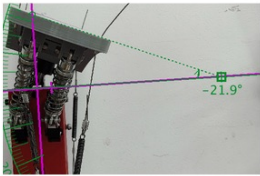


Figure 38: Inversion

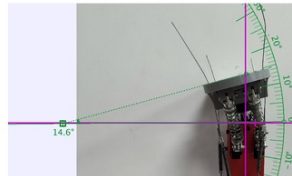
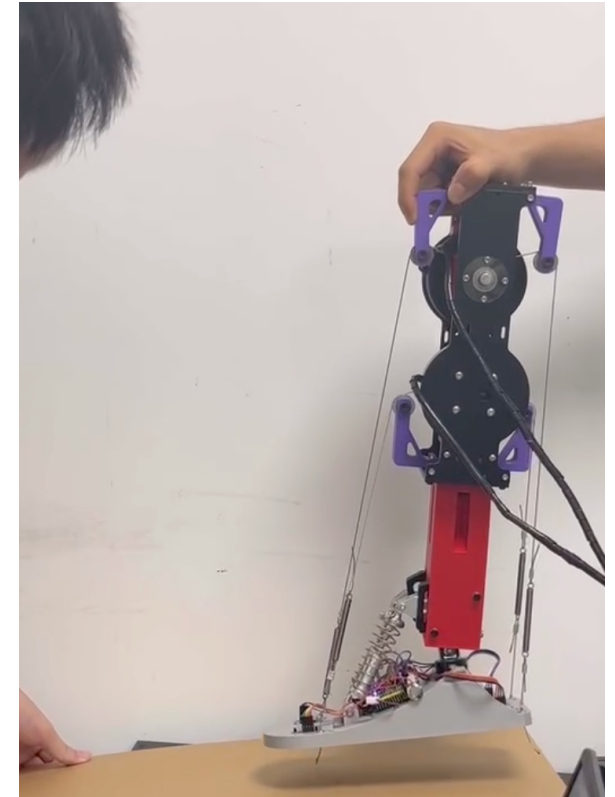
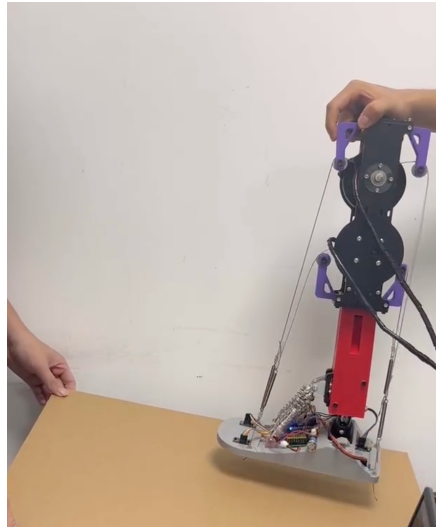
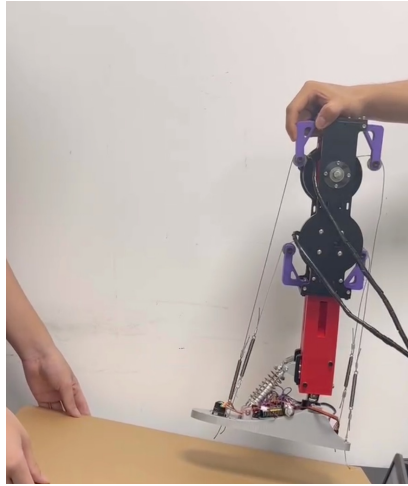


Figure 39: Eversion

- I use my hands to turn the motor such that the foot is in the extreme position. Even if the foot can still move further, it is unnecessary because the strings are too strained to be stable and safe.
- I take a picture of the lower part of the foot with the bottom face of the supporting leg perpendicular to the photo frame, and the bottom face of the foot perpendicular to the photo frame.
- I use Tracker to analyze the photo:
 - *Take the bottom of the supporting leg as the x-axis, since it is perpendicular to the photo frame, the bottom of the supporting leg is represented by a line.*
 - *Use the bottom of the foot as the other ray that locks the angle with the x-axis. The intersection between the line formed by the bottom of the foot and x-axis is the center of the protractor.*

Movement	Maximum Angle	Human's Maximum Angle	Percent Error
Plantar Flexion	33.7°	40°	16%
Dorsiflexion	24.1°	20°	20.5%
Inversion	21.9°	23°	4.7%
Eversion	14.6°	12°	22%



Self Adaptive Experiment

- The foot can adapt to the different inclinations of the ground.

Conclusion

- This bionic ankle has a 2 DoF, and a self-adaptive program to help it transverse through complex environments which search and rescue robots will benefit from.

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